

1. Explain why the Rankine cycle rather than Carnot cycle is used as a standard reference for the performance of steam plants.
2. Draw T-s diagram of Rankine cycle using dry-saturated steam and develop the equation for the Rankine cycle efficiency. Sketch a schematic diagram of a condensing steam power plant and show the sequence.
3. Discuss with the help of T-s diagram, the effect of the following variables on the efficiency and power output of a Rankine cycle
 - (a) Inlet pressure
 - (b) Inlet Temperature with inlet pressure maintained constant, and
 - (c) Condenser pressure.
4. In a Rankine cycle, the steam at inlet to turbine is saturated at a pressure of 30 bar and the exhaust pressure is 0.25 bar. Determine (i) The pump work, (ii) Turbine power, (iii) The Rankine efficiency, (iv) the condenser heat flow and (v) the dryness at the end of expansion. Assume flow rate of 10 kg/s.
5. A steam engine operates on ideal Carnot cycle using dry saturated steam at 17.5 bar. The exhaust takes place at 0.07 bar into a condenser. Assuming that the expansion and compression are isentropic and liquid enters the boiler as saturated liquid, find (a) the power developed by the engine if the steam consumption is 20 kg/min and (b) the efficiency of the operating cycle.
6. A boiler feed pump works on the (full admission) non-expansive cycle. Steam is supplied at 10 bar and dry saturated condition. The exhaust takes place at 1 bar. Draw the cycle of operation on p-v and T-s diagrams and find (a) the steam consumption per kW-hour. (b) Theoretical efficiency of the cycle. (c) Heat removed in the condenser per kg of steam. (d) If the feed pump supplies 100 kg of water per minute to the boiler, find the power required
7. Steam at 90 bar and 480°C is supplied to a steam turbine. The steam is reheated to its original temperature passing the steam through Reheater at 1 bar. The expansion after reheating takes place to condenser pressure of 0.07 bar. Find the efficiency of the reheat cycle and work output if the flow of steam is 5 kg/sec. Neglect the pressure loss in the system and assume the expansions through the turbines are isentropic.
8. Steam at a pressure of 100 bar and 500°C is supplied to a turbine. It comes out at 0.07 bar and 85% dryness. One stage of reheating is used. The net power output from the h turbines is 1600 kJ/kg. Determine the thermal efficiency of the plant.
9. The steam at 20 bar and 360°C expands in a turbine to a pressure of 0.08 bar. The steam coming out of turbine is condensed in a condenser to saturated liquid water. Then the pump feeds back the water (condensate) to the boiler. Assuming ideal processes, find out the cycle efficiency and power developed if the steam flow rate is 25 kg/sec.

***** **THE END** *****